Urban Heat Shade Dash

Find the coolest route through a hot campus.

Win condition: Best data-backed cool route.

THE SCIENCE YOU ARE USING

Urban heat and shade. Paved, sunlit surfaces get much hotter than shaded or planted ones. Shade from trees and buildings can drop surface temperature sharply over a few steps. Mapping these differences reveals cooler paths through a hot space.

Data-backed routing. You measure shaded and sunny surfaces, then design a cool corridor: a route that keeps people on the coolest surfaces. The best route is justified by your own temperature data.

HOW TO BUILD AND RUN IT

01 **Measure sun vs shade.** Record surface temperature in matched sunny and shaded spots.

02 **Map the hot spots.** Mark the hottest and coolest surfaces on your map.

03 **Design a cool corridor.** Plot a walking route that stays on the coolest surfaces.

04 **Justify with data.** Show the temperatures that make your route cooler than the direct path.

05 **Communicate the plan.** Present the route clearly so someone could follow it.

Safety: Check heat index. Walk, do not run. Allergy: Pollen sensitivity.

MATERIALS

Thermometers or IR thermometers	per team
Campus maps	per team
Clipboards	6

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______________________

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The ONE variable we are testing: ______________________

Baseline result: ______________________

After redesign: ______

DEFEND YOUR DESIGN

What evidence made you change your design? ______

If you ran it again, what is the ONE thing you would change? ______

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Cool route design	35
Data quality	25
Reasoning	20
Communication	20
Total	**100**